Name:_____

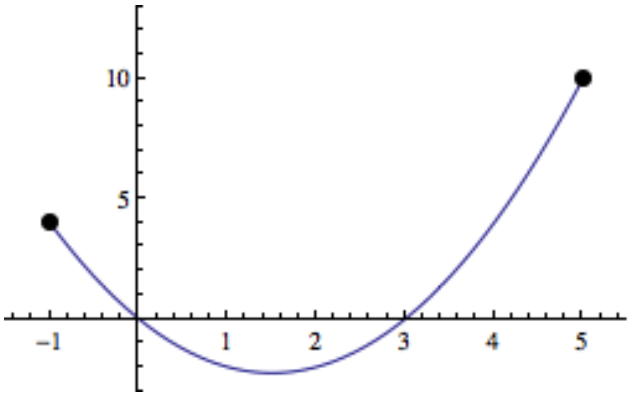
***For each graph in problems #1–#8, do the following.***

(a) Draw the secant line connecting the two endpoints given.

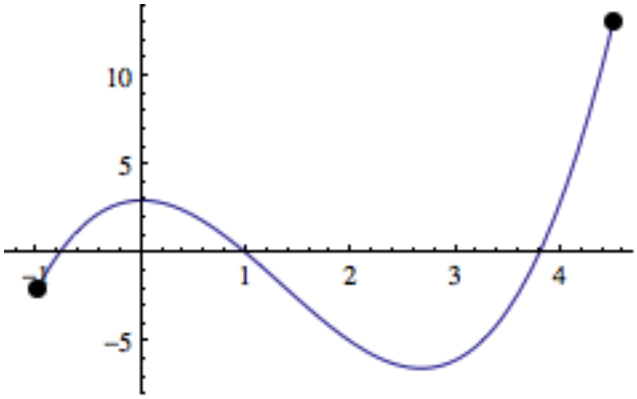
(b) Draw all tangent lines (if they exist) that are parallel to the secant line drawn in (a).

(c) Estimate the $x$-coordinates of any points of tangency for the tangent lines drawn in (b).

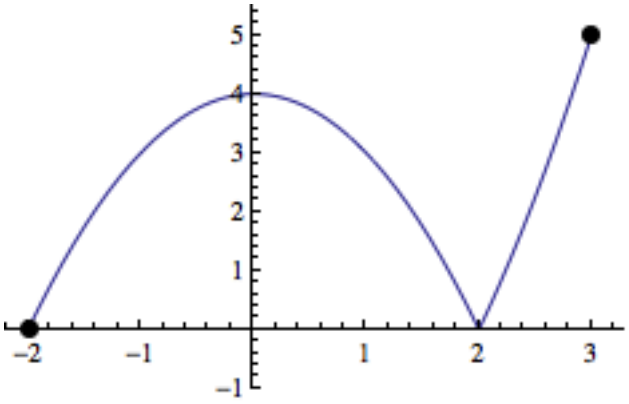
(1)



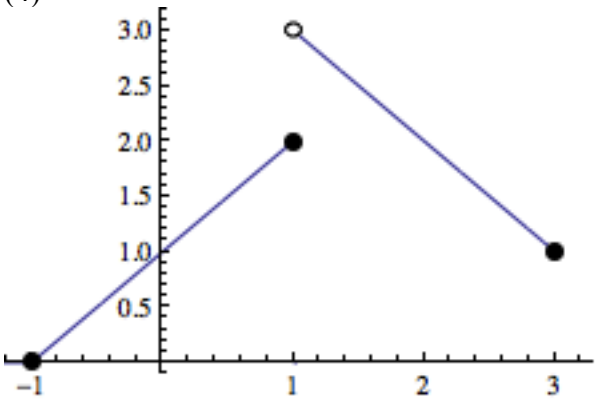

(2)



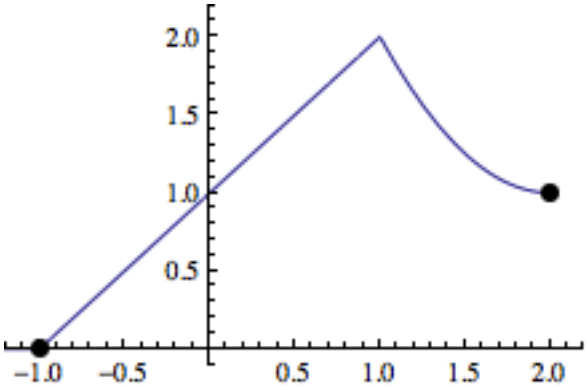

(3)



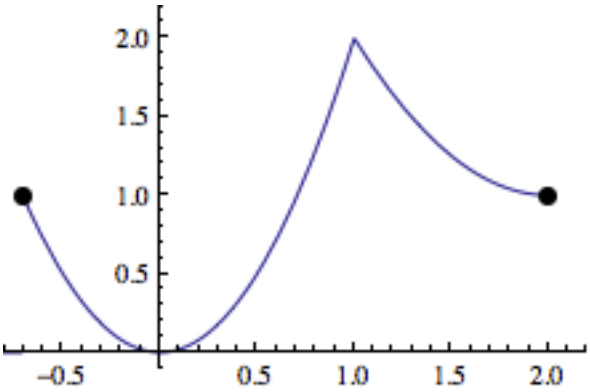

(4)



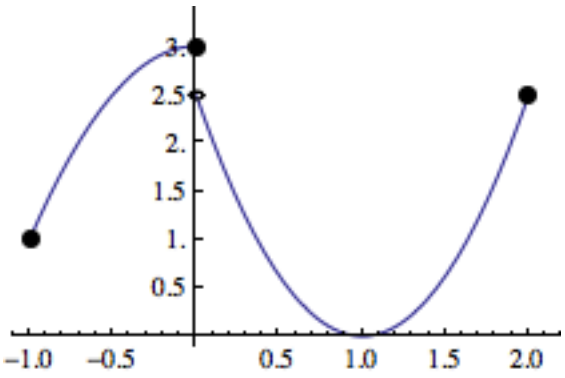

(5)



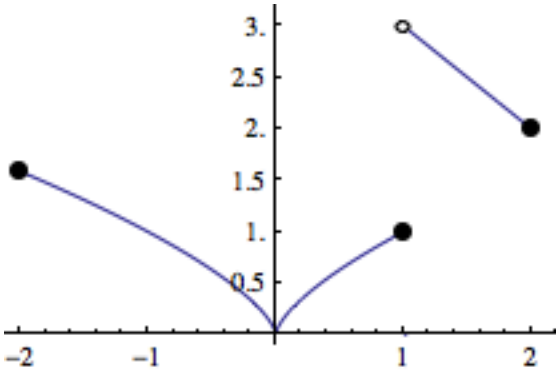

(6)



(7)




(8)




(9) For which of the graphs in problems #1–#8 were you able to draw at least one tangent line?

(10) Which of those graphs in problems #1–#8 are continuous?

(11) Which of those graphs in problems #1–#8 are differentiable?

(12) What characteristic(s) – continuity or differentiability – appears to guarantee that you can draw at least one tangent line parallel to the secant line connecting the given endpoints?

(13) In general, for a function $f$ on a closed interval $[a, b]$ and for $x = c$, the $x$-coordinate of the tangent line to the graph of $f$, write an equation that equates the slope of the secant line through the endpoints and the slope of the tangent line.